

Cognitive Fiber Dynamics: Entropic Descent and Modal Reflex in RSVP Field Space

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Abstract

We propose a computational and geometric theory of cognitive reflex within the Relativistic Scalar Vector Plenum (RSVP) framework, modeling semantic vectors in latent space as dynamic fields undergoing recursive transformations via fiber bundle dynamics and entropic descent. These transformations trigger activation thresholds in motor control (central pattern generators, CPGs) and phonological loops, establishing reflexive cognitive loops. By integrating neuroscience, provability logic, and geometric field theory, we develop a category-theoretic model of thought that is both simulation-friendly and formally rigorous. The model captures semantic stabilization, bifurcation, and recursive fixpoints, offering a novel synthesis for understanding cognition.

1 Introduction

The human brain orchestrates semantic transformations within high-dimensional latent spaces, dynamically shaping internal representations through sensory inputs and recursive feedback. Traditional symbolic and connectionist models struggle to capture the fluid, recursive nature of cognition, particularly its interplay with embodied action and phonological recursion. We introduce a geometric and field-theoretic framework, the Relativistic Scalar Vector Plenum (RSVP), to model cognition as the evolution of semantic vectors within a coupled scalar field Φ , vector flow field $\vec{v}$, and entropy field $\mathcal{S}$. These fields evolve under recursive entropy minimization and topological constraints, formalizing thought trajectories as geometric flows that stabilize, bifurcate, or trigger embodied loops via central pattern generators (CPGs) and phonological recursion.

Our contributions are threefold:

1. A formal model of thought as recursive transformations in RSVP fields, governed by entropic descent and fiber bundle dynamics.
2. A categorical framework mapping cognitive processes to modal fixpoints and Gödelian loops, bridging provability logic and neural dynamics.
3. A simulation-friendly paradigm for prototyping RSVP-based cognitive architectures, with applications to motor control and phonological processing.

This paper synthesizes insights from neuroscience [?], modal logic [?], and geometric field theory [?], offering a unified language for recursive cognition.

2 Background

2.1 RSVP Field Theory

The RSVP framework models cognition as a dynamic interplay of three fields: a scalar potential $\Phi(x, t)$, representing semantic amplitude; a vector flow $\vec{v}(x, t)$, encoding transformation dynamics; and an entropy field $\mathcal{S}[\Phi, \vec{v}]$, quantifying informational uncertainty. These fields evolve over a latent manifold $\mathcal{M}$, with dynamics governed by:

$$\frac{d\Phi}{dt} + \nabla \cdot (\Phi \vec{v}) = -\delta\mathcal{S}[\Phi, \vec{v}],$$

where $\delta\mathcal{S}$ is the functional derivative of the entropy field, driving the system toward minima of informational surprise, akin to predictive coding [?]. The vector field $\vec{v}$ acts as a connection, facilitating parallel transport of semantic content, while $\mathcal{S}$ induces curvature constraints.

2.2 Modal Logic and Fixpoints

Modal logic provides a formal lens for recursive cognition, with the necessity operator $\Box A$ representing provability or stabilization of a proposition A . Löb’s Theorem, $\Box(\Box A \rightarrow A) \rightarrow \Box A$, establishes conditions for self-referential fixpoints [?]. Gödelian loops, characterized by propositions $G \leftrightarrow \neg\Box G$, model non-terminating recursive processes [?]. These motifs map to cognitive dynamics, where fixpoints correspond to stable beliefs and Gödelian loops to oscillatory or divergent thought patterns. Recent discussions on memory and cognition further highlight the recursive nature of belief formation [?].

2.3 Neuroscientific Foundations

Central pattern generators (CPGs) are neural circuits producing rhythmic motor outputs, critical for locomotion and reflex [?]. The phonological loop, a component of working memory, sustains verbal rehearsal through recursive neural activation [?]. Predictive coding posits that the brain minimizes surprise by aligning internal models with sensory data, a process mirrored by entropic descent in RSVP [?]. These mechanisms underpin the reflexive loops modeled in our framework.

3 Geometric and Categorical Modeling

3.1 Latent Space as a Fiber Bundle

Semantic vectors reside in a latent manifold $\mathcal{M}$, lifted into a fiber bundle $\pi : E \rightarrow B$, where $B = \mathcal{M}$ is the base space, F is the fiber encoding complex amplitude and phase, and G is a structure group (e.g., $U(1)$ or $GL(n, \mathbb{C})$) encoding recursive transformations. The RSVP fields define a connection $\vec{v}$ and curvature $\mathcal{S}$, with dynamics:

$$D_t \Phi = \partial_t \Phi + \vec{v} \cdot \nabla \Phi,$$

where D_t is the covariant derivative ensuring parallel transport. This structure supports loops, bifurcations, and fixpoints, mirroring modal logic’s recursive forms.

3.2 Category-Theoretic Semantics

We define $\mathcal{C}_{\text{RSVP}}$ as the category of RSVP fields, with objects as field configurations $(\Phi, \vec{v}, \mathcal{S})$ and morphisms as transformations preserving semantic structure. The necessity operator $\Box : \mathcal{C}_{\text{RSVP}} \rightarrow \mathcal{C}_{\text{RSVP}}$ is a functor mapping fields to their recursive closures:

$$\Box A \cong \text{fix}(f : A \rightarrow A),$$

where f is an endomorphism satisfying $f(f(A)) \cong f(A)$. Gödelian loops correspond to objects outside $\Box$'s image, lacking global sections and exhibiting non-terminating dynamics.

4 Simulation Framework

4.1 Grid-based Discretization

We simulate RSVP dynamics on a 64×64 grid, initializing $\Phi_0(x)$ as a Gaussian semantic seed, $\vec{v}_0(x)$ as a randomized flow, and $\mathcal{S}_0$ via Kullback-Leibler divergence. The update rule is:

$$\Phi_{i,j}^{t+1} = \Phi_{i,j}^t - \Delta t \left(\nabla \cdot (\Phi^t \vec{v}^t)_{i,j} + \nabla \mathcal{S}_{i,j}^t \right).$$

Entropy smoothing uses a Gaussian kernel to stabilize field evolution.

4.2 Implementation of Modal Motifs

Löb-stable fixpoints are detected when $\Phi^t \rightarrow \Phi^*$ with $\nabla \mathcal{S} \cdot \vec{v} < \theta$. Gödelian loops manifest as persistent oscillations, where Φ^t fails to converge under $\Box(A) \rightarrow A$. We implement a stability criterion:

$$\text{Stability} = \|\Phi^{t+1} - \Phi^t\|_2 < \epsilon.$$

4.3 CPG and Reflex Modeling

CPG activation occurs when $\Phi(x, t)$ enters an attractor basin, defined by:

$$\nabla \mathcal{S} \cdot \vec{v} > \theta_{\text{CPG}},$$

triggering synthetic motor outputs (e.g., oscillatory patterns). Reflex inhibition arises in Gödelian cases, where oscillatory dynamics prevent threshold crossing.

5 Case Studies

5.1 Semantic Descent to Stable Gesture

A semantic vector encoding “reach” evolves under RSVP dynamics, stabilizing at Φ^* and triggering a CPG-driven arm movement. Simulation parameters: $\theta_{\text{CPG}} = 0.5$, $\Delta t = 0.01$.

5.2 Looping Phrase Generation

A phonological sequence “ABC” forms a recursive attractor, oscillating as “ABCABC...” with period $T \approx 0.1$ s, modeled as a fixpoint in Φ .

5.3 Reflex Inhibition via Gödelian Uncertainty

A divergent thought trajectory, akin to $G \leftrightarrow \neg \Box G$, fails to stabilize, inhibiting motor output due to unresolved semantic ambiguity.

6 Philosophical and Cognitive Implications

The RSVP framework posits cognition as a recursive interplay of geometric flows and modal fixpoints. Gödelian loops reflect cognitive uncertainty, manifesting as neural oscillations in decision stalemates [?]. The model suggests that free will emerges from bifurcations in RSVP dynamics, where choices align with entropic minima. This bridges ideomotor theory [?] and modal logic, offering a formal account of recursive belief formation and memory processes [?].

7 Related Work

Our model extends predictive coding [?] by incorporating geometric and categorical structures. Unlike field-theoretic models of cognition [?], RSVP emphasizes recursive fixpoints. Modal logic applications in AI [?] inspire our functorial treatment, while topos-theoretic neuroscience [?] informs our categorical framing. Recent discussions on memory dynamics further contextualize our recursive approach [?].

8 Conclusion

We propose that cognition, perception, and reflex are recursive transformations in RSVP fields, driven by entropic descent and modal closure. CPGs and phonological loops emerge from bifurcations and fixpoints, respectively. This framework unifies neuroscience, modal logic, and field theory, providing a simulation-ready and theoretically rigorous model of thought.

Future Work

- Prototype RSVP field lattices for real-time cognitive simulation.
- Formalize the topos of RSVP cognition.
- Explore quantum and stochastic RSVP modalities.
- Develop RSVP-based AI architectures.

A Field Update Equations

The full RSVP evolution equation is:

$$\frac{d\Phi}{dt} + \nabla \cdot (\Phi \vec{v}) = -\frac{\delta \mathcal{S}}{\delta \Phi}, \quad \mathcal{S}[\Phi, \vec{v}] = \int \Phi \log \frac{\Phi}{\Phi_0} d\mu.$$

The vector field evolves via:

$$\frac{d\vec{v}}{dt} = -\nabla \Phi + \kappa \nabla \times \vec{v},$$

where κ controls rotational dynamics.

B Category-Theoretic Proofs

Theorem 1. *The functor $\square : \mathcal{C}_{RSVP} \rightarrow \mathcal{C}_{RSVP}$ maps each field to its recursive closure, with $\square A \cong \text{fix}(f)$ for some endomorphism f .*

Proof. Define $f : A \rightarrow A$ as the morphism induced by RSVP dynamics. If $f(f(A)) \cong f(A)$, then $\square A$ is the stabilized field configuration, satisfying Löb’s condition [?]. $\square$

C Simulation Code Excerpts

```
import numpy as np
def rsvp_update(phi, v, S, dt=0.01, theta=0.5):
    grad_S = np.gradient(S)
    div_phi_v = np.divergence(phi * v)
    phi_new = phi - dt * (div_phi_v + grad_S)
    return phi_new if np.linalg.norm(phi_new - phi) < 1e-3 else None
```

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